Predicting Biology-related Library of Congress Subject Headings for Scholarly Articles

Nazmul Kazi, Nathaniel Lane & Indika Kahanda

Gianforte School of Computing, Montana State University, Bozeman, MT

Kahanda BioMedInfo Lab

Introduction

- Institutional repository (IR) managers need to catalog scholarly work using subject headings to facilitate retrieval of related articles.
  - Library of Congress Subject Headings (LCSH) has been widely adopted by large and small libraries around the world [5].
  - Manual cataloging is highly resource-consuming and covers limited number of articles per day.
  - IR items are growing rapidly and manual cataloging is becoming a challenge.
- Previous attempts to assign LCSH automatically depends on the presence of keywords [1,4] or similar words/phrases [3] but do not utilize any machine learning.
- In this work, we explore the feasibility of developing an automated pipeline for predicting LCSH using machine learning as a multi-label classification problem and our preliminary results indicates the effectiveness of our approach.

Methodology

Training Data:

- By scraping RAMP data of 27 institutional repositories [2].
- Metadata: title, abstract, article type and keywords.
- Each subject with at least 100 articles.
- Total 17,367 articles with 66 biology-related LCSH.

Decision Tree (Baseline):

- Lemmatization with stop word removal.
- TF-IDF vectorizer with word-based analyzer including both unigrams and bi-grams as features. Top 10,000 features only.

Artificial Neural Network:

- TF-IDF scores as features.
- It has three layers:

Input layer	:	2,251 nodes
A hidden layer	:	132 nodes
Output layer	:	66 nodes

Experimental Setup and Metrics:

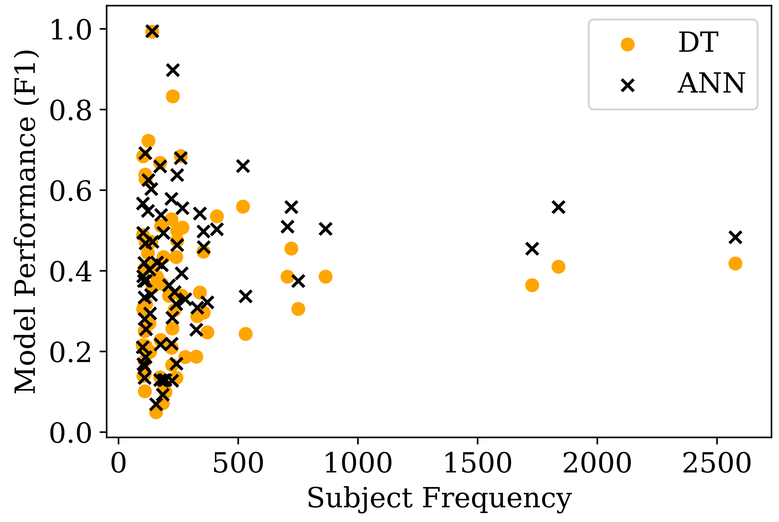
Evaluation:	5-times 5-fold stratified cross-validation
Performance Metrics:	Precision, Recall, and F-measure
Aggregation:	Macro-averaging

Results

Model performance per subject frequency range:

Subject frequency range	Number of unique subject headings	Decision Tree			Artificial Neural Network		
		P	R	F1	P	R	F1
[100, 200)	35	0.36	0.35	0.36	0.57	0.30	0.37
[200, 300)	15	0.40	0.39	0.39	0.57	0.35	0.42
[300, 400)	6	0.31	0.30	0.30	0.54	0.32	0.40
[400, 900)	7	0.41	0.41	0.41	0.58	0.43	0.49
[1700, 2600]	3	0.40	0.40	0.40	0.55	0.46	0.50
	Macro average:	0.38	0.37	0.37	0.56	0.37	0.44

Model performance over subject frequency:



Discussion

- ANN model outperforms DT model.
- DT model is seemingly immune to the frequency of subject.
- ANN model suggests a positive correlation between the performance and the frequency of subject.
- Best performed subject: *Commencement ceremonies* (both models).
- Most difficult subject: *Social psychology* (both models).
- Best performed document type: *Thesis* (both models).
- Most difficult document type: *Book* (both models).

Future Work

- A chain-classifier that predict subjects using previous predictions.
- Map articles using identifiers rather than names.
- Include hand-engineered features, MeSH terms and/or other metadata fields for training.
- Use algorithm that leverages the hierarchy information of LCSH.

Acknowledgements

Patrick OBrien and Kenning Arlitsch from Montana State University Library and Jonathan Wheeler from University of New Mexico.

References:

1. Aga, R. T., Wartena, C., and Franke-Maier, M. Automatic recognition and disambiguation of library of congress subject headings. In *International Conference on Theory and Practice of Digital Libraries* (2016), Springer, pp. 442-446.
2. OBrien, P., Arlitsch, K., Mixter, J., Wheeler, J., and Sterman, L. B. Ramp-the repository analytics and metrics portal. *Library Hi Tech* (2017).
3. Paynter, G. W. Developing practical automatic metadata assignment and evaluation tools for internet resources. In *Proceedings of the 5th ACM/IEEE-CS Joint Conference on Digital Libraries (JCDL'05)* (2005), IEEE, pp. 291-300.
4. Wartena, C., Brussee, R., and Slakhorst, W. Keyword extraction using word co-occurrence. In *2010 Workshops on Database and Expert Systems Applications* (2010), IEEE, pp. 54-58.
5. Work, J. A. Legislating librarianship. *The Political Librarian* 2, 2 (2016), 7.

Contact Info:

Nazmul Kazi	:	kazinazmul.hasan@montana.edu
Nathaniel Lane	:	nathaniel.lane@student.montana.edu
Indika Kahanda	:	indika.kahanda@montana.edu